

Functional sequences, series and spaces – 2

Series of functions

13 сентября 2026 года

Theoretical minimum.

Theorem 1 (Weierstrass' test). If $|u_n(x)| \leq c_n$ on X and $\sum c_n < \infty$, then $\sum u_n$ converges uniformly and absolutely on X . This is the only test that gives both at once, and it is useless exactly when the convergence is conditional.

Theorem 2 (Dirichlet and Abel for series of functions). If the partial sums of $\sum v_n(x)$ are uniformly bounded and $u_n(x) \downarrow 0$ uniformly, then $\sum u_n v_n$ converges uniformly. If $\sum v_n$ converges uniformly and $u_n(x)$ is monotone in n and uniformly bounded, the same conclusion holds. These are the tools for $\sum \frac{\sin nx}{n^\alpha}$ with $0 < \alpha \leq 1$, where Weierstrass has nothing to say.

Theorem 3 (term-by-term operations). A uniformly convergent series of continuous functions has a continuous sum and may be integrated term by term over a segment. For differentiation the hypothesis must be moved to the derivatives: if $\sum u_n(x_0)$ converges and $\sum u'_n$ converges uniformly, then $\sum u_n$ converges uniformly and $(\sum u_n)' = \sum u'_n$.

Fact 1 (how singularities are built). A series $\sum 2^{-n} \varphi(x - r_n)$ over an enumeration (r_n) of the rationals places a copy of the singularity of φ at every rational at once; a series $\sum \frac{\{nx\}}{n^2}$ places a jump at every rational with a controlled size. This is the standard machine for producing a function with a prescribed set of bad points.

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1. Prove the tests of Weierstrass, Dirichlet and Abel for series of functions. Then determine on which sets the series $\sum \frac{\sin nx}{n^\alpha}$ ($\alpha > 0$), $\sum \frac{(-1)^n}{n+x^2}$ and

- $\sum x^n(1-x)$ converge uniformly, and where the convergence is only pointwise.
2. Prove that $S(x) = \sum_{n \geq 1} \frac{1}{n^2 + x^2}$ is infinitely differentiable on $\mathbb{R}$ and find the asymptotics of S at infinity; then prove that $\sum \frac{\sin nx}{n}$ converges at every point but not uniformly on any neighbourhood of 0, and identify its sum.
 3. Let $R(x) = \sum_{n \geq 1} \frac{\{nx\}}{n^2}$, where $\{t\}$ is the fractional part. Prove that the series converges uniformly, that R is discontinuous exactly at the rationals, and that R is Riemann integrable on $[0, 1]$.
 4. (RMO) Let $f_0: [0, 1] \rightarrow \mathbb{R}_+$ be continuous and define $f_{n+1}(x) = \int_0^x \frac{dt}{1+f_n(t)}$ for all n and all $x \in [0, 1]$. Prove that $(f_n(x))$ converges for every $x \in [0, 1]$ and compute the limit. Is the convergence uniform?
 5. (CIIM 2024) Let $(a_n)_{n \geq 1}$ be a sequence of real numbers and define functions $(f_n)_{n \geq 0}$ by $f_0(x) = 1$ and $f_n(x) = \int_{a_n}^x f_{n-1}(t) dt$ for $n \geq 1$. Find all sequences (a_n) such that $f_n(0) = 0$ for all $n \geq 2$. (It is not required that $f_1(0) = 0$.)